Aditya Amodkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.23
Narayana Junior College Kalyan	Class XII (Maharashtra HSC Board)	2023	88.33%
All Saints High School	Class X (Central Board of Secondary Education)	2021	97.20%

WORK EXPERIENCE

Data Science & Analytics Intern | American Express

- **Reduced** manual effort and time **by ~67% (from 120 to 40 hours)** by automating the end-to-end Country Risk Assessment & Reporting workflow, from data acquisition and processing to country-wise risk-score calculation and report generation
- **Expanded** country risk coverage **7.5×,** from **26 to 195** UN-recognized countries, covering potential new markets, and enabling standardized risk assessment process across all countries using **17** economic, financial, and other country-level parameters
- **Built** an AI-enabled data pipeline using Power Query, API integrations, Excel, SQL and Custom GPTs, automating data retrieval across multiple sources, score calculations, and reporting to build a scalable and standardized country risk reporting tool
- **Collaborated** with risk and compliance stakeholders on reporting requirements to standardize and automate the workflow

PROJECTS

DevQuest, Multi-User Coding Arena

- **Developed** real-time collaborative coding platform with React and Socket.io for synchronized code editing, language switching
- **Engineered** an Express.js proxy integrating the JDoodle API for multi-language code execution in JavaScript, Python and C++
- **Architected** a modular full-stack app integrating Monaco Editor, WebSocket communication, Mongoose data management

SyncSpace - Real Time Collaborative Video Platform

- **Engineered** a real-time collaborative video platform synchronizing playback, navigation, and presence across users
- **Designed** server-authoritative conflict resolution using atomic Redis Lua scripts, sequence numbers, and stale-event rejection
- **Implemented** playback drift correction, reconnection recovery, and secure authentication; validated functionality through 83 tests

PriceMemory - Agentic AI E-commerce Intelligence Platform

- **Engineered** AI-driven pricing intelligence platform combining live product extraction, price tracking, and offer-condition analysis
- **Developed** Buy/Wait/Monitor recommendations using price trends, observation quality, and confidence-aware decision logic
- **Integrated** Gemini-powered investigation to summarize verified facts, risks, missing information, and actionable next checks

Customer Personality Analysis & Segmentation Model

- **Segmented 2,236** customers into 4 distinct personas using K-Means, revealing that Premium Loyalists **(23%)** generated **~52%** of total spend, enabling targeted retention and upselling

- **Evaluated 9** classification models, achieving **97.8%** accuracy for segment prediction with Logistic Regression and **0.913** ROC-AUC for campaign-response prediction using Gradient Boosting

Credit Card Fraud Detection Model

- Developed fraud classifiers on **284k+** credit-card transactions, comparing Logistic Regression, Decision Tree, Random Forest, and KNN using precision, recall, F1-score, and confusion matrices
- Achieved **~93%** accuracy with Random Forest, attaining **0.99** precision and **0.87** recall for fraudulent transactions, while Logistic Regression achieved **0.93** F1-score for the fraud class

Hangman Problem Solver - ML Hackathon

- **Developed a character-level Transformer ensemble with a train-only candidate-posterior hybrid**, improving Hangman win rate **to 52.2%** on a held-out 12K-word vocabulary through adaptive board-conditioned letter prediction and probability blending
- **Engineered and evaluated 6+ modeling strategies,** including GRU, Transformers, multi-task learning, candidate retrieval, and ensemble methods, by synthetic game-state generation and sequential simulation to optimize performance

SKILLS

Programming & Tools: C/C++, Python, JavaScript, TypeScript, SQL, HTML, CSS, Pandas, MongoDB, Redis, Power BI, DAX, Excel, React, Node.js, Express.js, Next.js, Socket.IO, REST APIs, Figma, Git, Github, Streamlit

Machine Learning & AI: Scikit-learn, Predictive Modeling, Feature Engineering, NLP, LLMs, RAG, Prompt Engineering

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, System Design

POSITIONS OF RESPONSIBILITY

Head Placement Coordinator, Civil Department | Office of Career Services, IIT Hyderabad

- **Leading** placement activities for **120+** department students, coordinating a team of **15** Placement Coordinators

Head of Consulting | Finance & Consulting Club, IIT Hyderabad

- **Led** a team of **18 members** to drive the Consulting domain across campus-wide consulting and product initiatives
- **Enabled** structured preparation for **800+ students** for management opportunities by **building IIT Hyderabad's first FCC Casebook**, covering consulting & product fundamentals, frameworks and **40+** campus interview cases

Public Relations & Marketing Coordinator | TEDxIITHyderabad

- **Doubled** Instagram following and generated **16k+ views** through audience-led content and multi-platform campaigns

EXTRACURRICULAR

Community Service: Volunteered with the National Service Scheme for social outreach and community service initiatives

Sports: Recreational badminton player, pursuing the sport for fitness and balance beyond academics